Despliegue local en Kubernetes con Minikube/Kind

Entorno de múltiples contenedores

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1. Objetivo

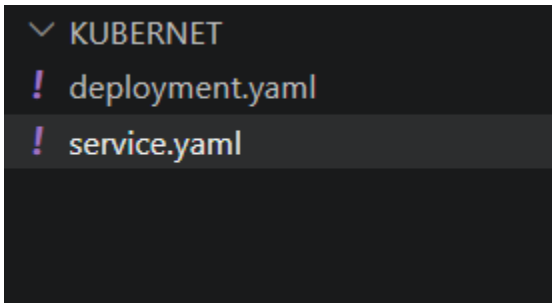
Validar el despliegue y exposición de una aplicación simple en Kubernetes usando Minikube como entorno local.

2. Entorno de trabajo

Campo	Valor
Plataforma	Kubernetes Local
Herramienta de edición	Visual Studio Code / Minikube
Directorio de trabajo	C:\Programacion\...\PROG 5\compose-practica
SO	Windows/Linux
Editor	VS Code
CLI	Kubetctl
Imagen usada	Ngix
Puerto expuesto	30080

3. Estructura del proyecto

Se creó la siguiente estructura de archivos dentro de la carpeta compose-practica:



4. Archivos del proyecto

4.1 instalar minkube

Dependencias Python necesarias para la aplicación web:

```
minikube start
```

4.2 Verificar clúster

```
kubectl get nodes
```

4.3 deployment.yaml

Imagen personalizada para el servicio web:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: web-app
  template:
    metadata:
      labels:
        app: web-app
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

4.4 service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: web-service
spec:
  type: NodePort
```

```
  selector:
    app: web-app
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
```

5. Ejecución del entorno

5.1 Construcción y arranque

```
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
```

5.2 Verificación

```
kubectl get pods
kubectl get svc
```

5.3 Acceso a la aplicación

```
minikube service web-service
```

6. Evidencia

```
PS C:\WINDOWS\system32> minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\WINDOWS\system32> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\WINDOWS\system32> minikube start --driver=docker
* minikube v1.38.1 en Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Descargando Kubernetes v1.35.1 ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 3.13 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 2.33 Mi
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Preparando Kubernetes v1.35.1 en Docker 29.2.1...
! Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Configurando CNI bridge CNI ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Complementos habilitados: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\WINDOWS\system32>
```

```
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5> kubectl apply -f deployme
deployment.apps/web-app created
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5> kubectl apply -f service.
service/web-service created
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5> kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
web-app   0/2     2            0           10s
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5> kubectl get pods
NAME                       READY   STATUS              RESTARTS   AGE
web-app-85f95dbb6b-drlhs   0/1     ContainerCreating   0          14s
web-app-85f95dbb6b-j8r7s   0/1     ContainerCreating   0          14s
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5> kubectl get svc
NAME          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes    ClusterIP   10.96.0.1        <none>        443/TCP        10m
web-service   NodePort    10.109.148.112   <none>        80:30080/TCP   12s
```

```
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\WINDOWS\system32> cd "C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5\KUBERNET"
PS C:\Programacion\POGRAMACION\UNIVERSIDAD UIP\PROG 5\KUBERNET> minikube service web-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-service	80	http://192.168.49.2:30080

```
* Starting tunnel for service web-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-service		http://127.0.0.1:53920

```
* Opening service default/web-service in default browser...
! Porque estás usando controlador Docker en windows, la terminal debe abrirse para ejecutarlo.
```

127.0.0.1:53920

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.